Directed Cyclic Networks
Phase 1B: Adversarial Novelty Audit

24 September 2026

Research audit with explicit access gaps. No proposed project theorem is proved or numerically tested. The canonical models and original audit are preserved.

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1 Phase 1B: adversarial novelty audit

1.1 Verdict and limits of closure

The original eight targets do not presently support eight novelty claims. Four basic claims are unsafe as principal contributions: Fourier diagonalization, the fixed-symbol large-network limit, Fourier inversion for static response, and generic perturbation bounds. Mode selection and the first Lyapunov coefficient require substantial sharpening. The matched comparison is a potentially useful specialization. Priority for the demand-modulated coupling law remains unresolved. No target receives GREEN in this audit.

This is a source-based assessment, not verification of the blueprint's mathematics. The canonical demand-modulated model (DM), matched bilinear comparator (MB), positive parameter domain, indexing, and terminology remain unchanged. All P01–P18 files remain TODO. Neither an unsuccessful search nor a different displayed equation establishes novelty. No proof or numerical experiment was performed. Recommendations below are proposals for subsequent author decisions.

The six supplied PDFs and the original Phase 1 audit remain the baseline. This revision adds English and Chinese searches, backward citation tracing, and stronger generic counterexamples to the novelty narrative. The search cutoff is 24 September 2026. Access restrictions prevent full closure of several five-element sources. Their missing equations are recorded as missing, not reconstructed from abstracts. A separate archival research track is neither created nor recommended.

1.2 Evidence convention and source locations

S01–S14 retain their Phase 1 identifiers. Page numbers for S01–S04 and S08 are those of the inspected author manuscripts, not silently converted journal pages. New records specify their version. "Proves" below means that an argument is present in the source, not that this audit independently certifies every line. "Stronger" refers only to the stated overlapping component and its assumptions; it does not mean that a different vector field contains DM. Search-index excerpts of primary PDFs are explicitly distinguished from full-text inspection. The files `additional_sources_needed.md` and `audits/phase1b_search_log.md` document the unresolved frontier.

1.2.1 Reassessment of the original six papers

S01: Jaramillo, Mrad and Stepien (2023).
[11]; DOI 10.1063/5.0138150. In arXiv:2210.04342v2, §III, p. 3, (5)–(6) define the three-species May–Leonard system with linear mutation. §IV.B, p. 5, Theorem 1 and (7)–(8) treat its circulant spectrum; §IV.C, pp. 6–7, Theorem 2 and (15)–(25) give a Hopf/Lyapunov calculation, including Fourier vectors in (22)–(23). The competition coefficients satisfy $\alpha, \beta > 0$; the stability discussion separates $\mu \geq 1/6$ from $0 \leq \mu < 1/6$. The Hopf calculation uses the critical line (11) and a nonzero frequency; one must exclude its zero-frequency coincidence. The proofs concern that three-species polynomial system. Relation: orthogonal model, overlapping Hopf calculation method; not an arbitrary-(n, q) DM theorem.

S02: Pais, Caicedo-Núñez and Leonard (2012).
[23]; DOI 10.1137/120878537. The supplied 30-page manuscript uses a replicator–mutator model on a simplex, §2, (2.2), pp. 3–5. The circulant family in §4, (4.1), p. 10, has $N \geq 3$, the specified matrix entries and mutation structure; its parameters satisfy $\alpha, \beta \in [0, 1)$ and $\alpha + \beta > 0$. §4.1, pp. 11–13, (4.2)–(4.4), Lemmas 4.2/4.4 and Theorem 4.5 give mode/Hopf results. Lemma 4.6, p. 13, and Appendix D, pp. 26–28, (D.1)–(D.4), reduce the Lyapunov coefficient by Fourier structure, including

cubic terms. Appendix B, pp. 24–25, records the general formula. Relation: different state space and kinetics; strong direct precedent for the proposed method, including the cubic part.

S03: van Kekem and Sterk (2018): the closest theorem-program precedent.
[27]; DOI 10.1016/j.physd.2017.11.008. In the 55-page accepted manuscript, (1), p. 1, is Lorenz–96, with scalar forcing F and a homogeneous equilibrium. §2.1, pp. 8–9, (3)–(6) gives a circulant spectrum with real dispersion involving $\cos\theta - \cos 2\theta$. Proposition 5, pp. 11–12, selects the first unstable mode for $F > 0$, $n \geq 4$, maximizing over $0 < l < n/3$; its bounds have an $n = 7$ exception and $n = 12$ has a two-pair tie. Theorem 2, p. 10, treats simultaneous crossings. Theorem 4, p. 11, gives an explicit first Lyapunov coefficient and criticality information for $n \geq 4$, $0 < l < n/2$, $l \neq n/3$, when that pair is the only one crossing at $F_H(l, n)$. Proposition 6, §2.3, p. 13, gives the selected-wave limit $n/l_1(n) \to 2\pi/\arccos(1/4)$, with its argument on p. 13. The Lyapunov/selection proofs appear in §§4.3–4.4, pp. 38–48. These are proved results for Lorenz–96, not DM. Relation: orthogonal equation, but substantially the same proposed sequence of spectral selection, large-n asymptotics and Hopf calculation. Changing coefficients in a comparable dispersion relation is insufficient by itself to establish a substantive new theorem program.

S04: Wei et al. (2021).
[28]; DOI 10.1016/j.jfranklin.2020.11.011. The supplied 18-page manuscript defines the five-dimensional dynamo at (2.1), p. 3; Theorems 2.1–2.2, pp. 4–5, discuss Hopf points using reported numerical quantities. Theorems 3.1–3.2, §3, pp. 5–7, concern the specified controlled/observer error system. Five coordinates do not make this a scalar circulant ring. Relation: orthogonal to the DM coupling law and passive parameter-robustness bounds. Neither its control conclusions nor its reported decimals establish our claims.

S05: Lin and Han (2020): exact comparator overlap at five nodes.
[15]; DOI 10.23919/CCC50068.2020.9188438. §2.1, p. 778, (2) is the six-term homogeneous five-element polynomial model with nonnegative coefficients. Our $\mathrm{MB}_{5,2}$ is its exact subfamily under the coefficient assignment $(a_L, b_L, c_L, d_L, g_L, h_L) = (a, k, c, h, 0, g)$. §2.2, p. 778, discusses positivity; §2.2.1, Case 4, p. 779, (5)–(7), addresses homogeneous equilibria. Numerical examples are in §§2.2.2–2.2.3, pp. 779–781. Relation: the source model is stronger in parameter count at $n = 5$; MB is identical on that subfamily. This is a source-equation identification, not a new analysis of MB. The comparison with DM still requires a distinct result. References on p. 781 lead to earlier Chinese and English five-element models.

S06: Nian and Li (2025).
[22]; DOI 10.1186/s13662-025-03909-9. In the publisher PDF, §2.2, pp. 5–6, (1) defines a heterogeneous five-variable linear/quartic model; (3), p. 7, is its Jacobian. §§3–4, pp. 8–13, report fitted parameters and dynamical computations. Relation: orthogonal coupling law, overlapping five-element motivation. Reported simulations do not prove arbitrary-ring statements or validate the biological interpretation of DM.

1.2.2 Generic theory that defeats broad novelty claims

S07: Gray (2006).
[9]; author PDF, §3.1, printed pp. 32–34, (3.7)–(3.13); §3.2, Theorem 3.1, printed pp. 34–35. Finite circulant matrices have a unitary Fourier decomposition; inverse formulas require nonzero eigenvalues. These results are established in the source. Relation: strictly more general linear algebra than our homogeneous Jacobian, once that Jacobian has been correctly derived. Spectrum, normality and inverse Fourier sums are supporting applications, not new general theorems.

S08: Kerin and Engler, now identified in its published form (2022).
[12]; DOI 10.3934/dcdsb.2021064. The inspected version remains arXiv:2005.07767v2: §3.2, pp. 11–12, (27)–(29), Lemma 1; Appendix A, p. 27, Proposition 4, (48)–(49). Assumptions include a circulant linear part, an equivariant quadratic nonlinearity and a simple nonresonant Hopf setting with the displayed inverses defined. The scalar Fourier reduction is proved for this class. Relation: broader quadratic class, but not the full rational DM nonlinearity. §5, pp. 21–24, (40), Propositions 1–3, allows inhomogeneous coefficients under energy-preserving quadratic and positive damping hypotheses; those hypotheses must not be imported into DM. The new bibliography entry adds the journal record while preserving the original preprint entry.

S09 and S22: equivariant Hopf and phase patterns.
Stewart [25], arXiv:2403.15814v1, §3, Theorem 3.7, and §§5.2–5.4, (16)–(17), Theorems 5.5/5.7 and Corollary 5.6, supplies a ring-specific synthesis. The original research source Filipski–Golubitsky [7], DOI 10.1137/090756582, §3, p. 285, Theorem 3.1, and §5, p. 288, Theorem 5.1, proves abelian equivariant Hopf symmetry restrictions/realization. The hypotheses concern smooth generic equivariant bifurcation, a critical representation and isotropy/kernel conditions; the realization theorem is not existence in every particular vector field. Relation: general symmetry theory subsumes our conditional rotating-wave conclusion. DM must still supply the simple imaginary pair, crossing, nondegeneracy and all-mode stability conditions. Do not copy the phase expression in Stewart v1 (15) without reconciling it with the surrounding convention.

S10–S11: matrix perturbation and threshold sensitivity.
Embree–Trefethen [6], DOI 10.1137/S1064827500373012, §1, pp. 583–584, Theorem 4ε, p. 585, and the normal-matrix discussion on p. 586, relates Euclidean perturbations, resolvents and pseudospectra. Its normal-matrix case is broader than our stable homogeneous linearization. Greenbaum–Li–Overton [10], DOI 10.1137/19M124784X, author PDF §2, p. 4, Assumption 1, Theorem 1 and (1), proves the derivative of a simple eigenvalue for an analytic matrix family with normalized left/right eigenvectors. Relation: the general derivative is stronger than its DM application. Differentiating a simple transverse threshold is routine after the moving equilibrium has been included.

S19: Benzi and Razouk (2007), a stronger localization baseline.
[3]; published ETNA PDF. §3.5, p. 24, Theorem 3.4 and (3.7), proves decay in directed graph distance for diagonalizable $A = XDX^{-1}$ whose spectrum lies in a continuum F, with f analytic on the larger domain $I(C_{R_0})$. The bound contains $\kappa(X)$. §3.7, p. 27, Theorem 3.7 and (3.10), supplies explicit constants under additional geometric hypotheses. Relation: stronger generic matrix-function theory; using $f(z) = 1/z$ requires the analytic region to avoid zero. Uniform-in-n claims need uniform hypotheses, not just invertibility at each size. A cyclic graph distance must respect wraparound; ordinary band distance need not. This closes the generic decay-method gap even though the full S12 Demko–Moss–Smith text [4] remains only partly accessible.

S20: Sun, Bollt and Nishikawa (2009).
[26]; DOI 10.1209/0295-5075/85/60011. In arXiv:0811.0649v1, p. 2, (1) is a uniformly coupled near-identical network; pp. 4–5, (6)–(9) diagonalize its inhomogeneous variational equation when the modified Laplacian is diagonalizable. Pp. 6–7, (12)–(15), derive a forced-response bound under an exponential transition bound and bounded mismatch forcing. These are linearized conclusions; replacement of the mean trajectory by an isolated unit is explicitly conjectural on p. 6. Relation: overlapping method, different coupling class. It does not prove nonlinear DM persistence or a global robustness radius.

S21: Qiu et al. (1995).
[24]; DOI 10.1016/0005-1098(95)00024-Q. §1, pp. 879–880, (1)–(3), treats stable $A + BDC$ with fixed real A, B, C and a full real or complex perturbation block D measured in the spectral norm. The unnumbered Theorem, p. 886, (20), is the real structured-singular-value formula, with proof in §2. Relation: stronger exact linear robustness theory, but its full-block uncertainty is not arbitrary repeated/diagonal parameter uncertainty. It does not include a nonlinear equilibrium's displacement automatically. Backward reference: Hinrichsen–Pritchard (1986), DOI 10.1016/0167-6911(86)90094-0; its full text was not inspected here.

S23: Asllani et al. (2014).
[1]; DOI 10.1038/ncomms5517. In the 28-page arXiv:1402.0760 manuscript, §II, pp. 8–10, (6)–(11), develops directed-network dispersion and an instability condition. Assumptions include a stable two-species reactive equilibrium and a complete Laplacian eigenbasis; p. 8, footnote 2, requires balance for the Fickian transport convention to preserve homogeneity. The linear instability analysis is analytic; subsequent travelling patterns are illustrated numerically. Relation: broader graph setting but different multi-species diffusion kinetics, not exact subsumption of DM selection. Directed dispersion and pattern formation are already established subjects.

S24: Montanari, Zanin and Motter (2026): recent priority warning.
[21]; arXiv:2609.25226v1, submitted 21 September 2026. Supplement §S3, Assumptions 1–2, requires diagonalizable L, $N \geq 3$, nonnegative real parts of its eigenvalues, and nonnegative damping. §S3.2, Theorem 4, for (3)–(4), assumes non-Hermitian circulant L and differentiability at the homogeneous optimum; it proves this optimum is not locally optimal among heterogeneous damping choices. Equations (S49)–(S54) use second derivatives and Fourier mode sums. §S4 separates proved results from a mode-mixing conjecture. Relation: different second-order network model, substantial overlap with broad heterogeneity-improves-stability or mode-mixing narratives. This preprint has not been independently validated and is not evidence that every perturbation stabilizes DM.

1.2.3 Five-element and biochemical priority search

S13–S16: the Lin–Shi–Han enzymatic sequence is the main access gap.
S13 [18], DOI 10.23919/CCC52363.2021.9550151, has pages 6410–6415, now confirmed by S14 reference 37. S15 [19], DOI 10.23919/CCC55666.2022.9901543, pp. 5735–5740, and S16 [16], DOI 10.23919/CCC58697.2023.10239797, pp. 6823–6828, are identified in S14 references 38–39. S14 [17], DOI 10.1007/s11071-024-09374-5, publisher abstract and references, studies enzymatic regulatory network topologies and the role of five nodes. Complete equations/supplement were not retrieved. Exact assumptions, equation identity, and theorem overlap therefore remain GRAY for this sequence. An indexed author upload under the 2022 title contains a later uncorrected proof: versions must not be conflated. No theorem-level adverse or favorable priority conclusion is based on that upload.

S17: Zhang, Xia and Wang (2020).
[29]; DOI 10.1088/1674-1056/abc2bf. Publisher-PDF indexed text at §5.1, p. 120504-5, (1), specifies five nodes with production formed from Hill activation and Hill repression factors, and linear degradation. The Hill exponents are not network size. This is a definite prior rational/saturating Wu-Xing coupling construction; it differs from the displayed DM law. The publisher abstract reports a landscape/flux investigation. Access is partial: the equation passage and metadata were inspected, not the full numerical argument. Relation: orthogonal exact law, direct overlap with any

broad claim to introduce saturation into five-element networks. Reference 11, p. 120504-11, leads to Yang et al. (1995), requiring retrieval.

S18: Ma et al. (2009), a backward biochemical source.
[20]; DOI 10.1016/j.cell.2009.06.013. Experimental Procedures, "Equations of the Circuit", p. 772, (7), uses enzyme activities normalized to fixed total concentration one, positive kinetic constants, and source activity multiplied by saturating recipient activation/deactivation factors. Its three-node topology search and associated adaptation analysis concern that biochemical class. Relation: recipient-state modulation is already a modeling ingredient; the factors are not the DM fractions. This prevents equating a new name with a new coupling principle. A coordinate or parameter equivalence to DM would need actual verification.

S25: Ganikhodjaev, Pah and Rozikov (2019).
[8]; arXiv:1908.07740. The model is a discrete quadratic stochastic operator on a simplex, not an ODE. §4.3, p. 11, Theorem 1, and §4.5, pp. 13–14, Proposition 7/Theorem 4, contain dynamical results for the specified parameter/operator class. Relation: orthogonal to DM and continuous-time MB; relevant to prior mathematical use of the five-element graph, not evidence for an identical differential law.

S26–S29 and older Chinese sources: explicitly unresolved.
Ding–Wang (2008) [5], DOI 10.1007/s11726-008-0269-8, pp. 269–270: publisher abstract/reference list inspected, equations unavailable. Its references lead to Li (1987) [14], DOI 10.1088/0031-8949/36/6/015, pp. 966–969, and Zhang–Han (1998), pp. 130–135. Lin–Han (2020), p. 781, also points to Feng–Niu–Wang (2003), pp. 71–74, Wang's 2008 thesis, Bellavite et al. (1998) [2], and Kim et al. (2014) [13]. Abstracts/metadata are insufficient to exclude transformed rational or bilinear models. No missing theorem number or equation locator is invented. The retrieval list states exactly what must be inspected before T1 can be cleared.

1.3 What a conditional Hopf theorem would and would not add

Known theory supplies the local bifurcation mechanism and permissible spatiotemporal symmetry. Fourier mode arithmetic and use of the zero and second harmonics in a Lyapunov calculation also have explicit precedents (S02, S08). These are not separate discoveries. The still-unproved DM content is an admissible positive equilibrium branch, the first crossing on a specified path, exclusion of simultaneous critical modes and zero frequency, transversality including equilibrium motion, and a rigorously certified nonzero Lyapunov coefficient. An open parameter region with a qualitative MB contrast would make these checks serve one mathematical question. A list of conditional formulas with no certified realizability would not.

Neither a maximum real part at one parameter value nor a tied pair of conjugate modes proves a first Hopf bifurcation. Distinct critical pairs, the real Nyquist mode when n is even, and modular harmonic coincidences must be separated. Equivariant theory does not supply orbital stability without the relevant remaining spectrum and criticality conditions. Exact nonlinear phase symmetry must be distinguished from a leading sinusoidal approximation; its sign depends on the frozen shift convention.

1.4 Assessment of the four proposed packages

A: spectrum + step-two selection + large-network limit.
Likely significance in its basic form: modest. Diagonalization and continuous/discrete maximization

are largely generic; S03 is a particularly close precedent. The coefficient-to-parameter map and realizable mode regions would be model-specific. Referee objection: a quadratic optimization exercise appended to a new equation. Needed addition: a complete admissible parameter classification tied to a qualitative dynamical/comparator contrast, with exceptional sizes and ties treated. A limit alone does not cure this.

B: selection + rigorous Hopf + rotating-wave branch.
Potentially coherent, but generic Hopf and equivariance constitute much of the machinery. The useful specific content would be a certified first instability and criticality phase diagram on positive branches. Referee objection: S02/S03 already perform an analogous program. Needed addition: analytic realizability on nonempty open parameter regions, a meaningful MB contrast, and rigorous criticality. Merely finding an imaginary pair and invoking symmetry is insufficient.

C: selection + explicit Lyapunov coefficient + structured heterogeneity.
Potentially stronger if the last component explains a new mechanism. The tensor reduction, inverse Fourier sum, and first-order sensitivity are mostly known. DM derivatives and sharp parameter-dependent response could be specific. Referee objection: three routine calculations with little interaction. Needed addition: a sharp structured result tied to the selected mode or its nonlinear branch, with a controlled nonlinear remainder and a quantified gain over the generic baselines S19/S21. An assertion that disorder can help is insufficient in light of S24.

D: all of B and C.
Largest scope, not automatically greatest significance. The model-specific part would have to link wave selection, branch criticality and a meaningful heterogeneity mechanism. Referee objection: overextended catalogue with the difficult assertions left conditional. Needed addition: one unifying theorem and a clear reason the heterogeneity result belongs to the main argument. More standard lemmas do not replace such a result.

Recommended smallest package. Pursue a strengthened B with P13 criticality and P08 comparison as essential parts: an exact, realizable $q = 2$ selection/first-Hopf classification that explains a qualitative difference between DM and MB. Keep P04 as a supporting general lemma, P06 as a useful quantitative corollary only if it clarifies the mechanism, and P14 as the symmetry conclusion. Do not make the full P15–P18 extension a prerequisite for this core. If no such model-specific contrast or sign classification survives subsequent work, reconsider the contribution before expanding the theorem count. This recommendation is conditional, not a prediction of journal acceptance or a claim that the proposed strengthened results are true.

1.5 Recommended scope of the generalization

Retain arbitrary (n, q) for the model definition and general spectral background. Make $q = 2$ the substantive setting. At the level of a candidate symbol using the first and second spatial harmonics, the standard identity for $\cos 2\theta$ explains the quadratic structure; this observation is not a derivation of the DM coefficients. For other q, the algebra and competing modes need separate treatment. No complete arbitrary-q selection theorem is recommended now.

Varying n permits more mode classes, parity effects and possible mode competition unavailable at five nodes. These are motivations for the generalization, not verified DM phenomena. Show that the admissible parameters actually realize relevant changes; relabeling a sampled maximizer is not enough. Large-n work is useful when it predicts wavelength and finite-size threshold error under a

fixed, explicit family. It is weak if it only replaces a grid by a continuum or changes parameters with n without uniform control.

Present $n = 5, q = 2$ as the motivating example and a checkable specialization, with prior Wu-Xing models credited in the main audit. Golden-ratio trigonometric constants are not intrinsic novelty. The condition $\gcd(n, q) = 1$ concerns the step-q subcycle; it is not necessary for the full network's connectivity when the step-one edge is present. No restriction of the canonical parameter domain or alteration of the coupling law is justified solely by this novelty audit.

1.6 Closure gates before claiming a substantive contribution

The eight target decisions are in the accompanying revised matrix. Module dispositions and precise proposed strengthenings are in `novelty_decisions.md`. Priority for T1 remains blocked by the enzymatic sequence and older source equations. This does not prevent later independent mathematics if explicitly authorized; it prevents a responsible priority claim. Branch/path, comparison calibration, heterogeneity assignment and uncertainty norm also remain open choices. No result has been promoted to VERIFIED or AUDITED by this literature work.

2 Revised novelty matrix: eight adversarial decisions

This matrix replaces the original matrix only for current planning; the original file remains preserved. Exact source assumptions, locations and access limitations are recorded in Section 1. RED means known, routine or unsafe as novelty; ORANGE means substantial overlap requiring strengthening; YELLOW means a potentially new but modest specialization; GREEN would mean an apparently substantive new result; GRAY means undetermined. No GREEN finding is warranted. "Not located" is never a positive novelty argument.

ID	Target and owners	Verdict	Strongest adverse evidence / present use
T1	Demand-modulated model; 00	GRAY	S13–S18 and older Wu-Xing sources. Coupling priority unresolved; topology and saturation are not new principles.
T2	Arbitrary-(n, q) spectrum; P03–P04	RED	S07 general circulant theory, S09 ring formulation. Supporting calculation after the Jacobian is derived.
T3	$q = 2$ quadratic selection; P05	ORANGE	S03 Proposition 5 and its two-harmonic dispersion. Need realizable parameter regions and dynamical consequences.
T4	Large-n selected mode; P06	RED	S03 Proposition 6; fixed-symbol grid maximization is routine. Retain only with useful finite-size/threshold information.
T5	Matched bilinear comparison; P08	YELLOW	S05 contains MB at five nodes exactly. Potential novelty is the qualified DM–MB comparison, not MB itself.
T6	Explicit Lyapunov reduction; P13	ORANGE	S02 Appendix D (including cubic terms), S08 Lemma 1. Known method; model-specific sign information may matter.
T7	Heterogeneous Fourier Green function; P15–P16	RED	S07 inverse formula and S19 decay theorem. An inverse sum is not a localization theorem.
T8	Robustness bounds; P17–P18	RED	S10–S11, S20–S21. Generic gap and derivative formulas are known; sharp structured nonlinear bounds remain a separate proposal.

2.1 T1: demand-modulated model — GRAY

Exact published statement?

Undetermined. No inspected source has been established as equation-identical to DM; the highest-risk enzymatic papers lack accessible complete equations. This is a gap, not a novelty finding.

Immediate corollary?

A modeling definition is not a theorem corollary. It may nevertheless be a reparameterization, restriction or rearrangement of an existing kinetic family; that comparison is unfinished.

Routine supporting specialization?
The cyclic/five-element topology, activation/restriction motifs and saturation alone are known ingredients.

Known method, new closed form?
Possibly a new combination of known kinetic factors; neither provenance nor mathematical value is settled.

Substantive model-specific novelty?
Not established. "Demand-modulated" is the frozen descriptive name, not evidence of a new physical law.

Likely referee response?
Ask why these fractions are appropriate and what consequences they have beyond existing biochemical laws.

Required sharpening.
Complete the S13–S16 equation comparison and show a specific dynamical contrast under a transparent MB calibration. Do not rename the term as standard Holling type II or use biomedical interpretation as mathematical evidence.

2.2 T2: arbitrary-(n, q) Fourier spectrum — RED

Exact published statement?
The general diagonalization theorem is published (S07). The displayed DM coefficients have not been verified here.

Immediate corollary?
Yes, conditional on establishing that the actual homogeneous Jacobian is the claimed circulant polynomial in the shift.

Routine supporting specialization?
Yes: spectrum, normality and the modal Hurwitz test belong in the supporting analysis.

Known method, new closed form?
The substitution may produce a model-specific expression, not a new spectral method.

Substantive model-specific novelty?
No on the present formulation.

Likely referee response?
Fourier diagonalization of a circulant is standard; arbitrary offsets do not remove that objection.

Required sharpening.
Use this lemma to support P05/P08/P11. A genuinely new classification for other q would be separate work and is not needed in the recommended core.

2.3 T3: exact quadratic mode selection — ORANGE

Exact published statement?
Not verified for the DM coefficient family. S03 already contains a close two-harmonic selection problem.

Immediate corollary?
The basic quadratic maximization is an elementary consequence of a verified quadratic symbol; the admissible DM parameter classification and first-instability interpretation are additional tasks.

Routine supporting specialization?
A nearest-cosine rule alone is routine and should not be presented as a major discovery.

Known method, new closed form?
Yes, potentially. The coefficient map and its admissible regions may be a new specialization.

Substantive model-specific novelty?
A candidate only if mode regions, their realizability and a consequential DM–MB contrast are established.

Likely referee response?
Why is optimizing this quadratic materially different from the Lorenz–96 example?

Required sharpening.
P05 proposal S05 in the decisions file: classify all admissible maximizing mode pairs, ties and exceptional real modes, then connect the classification to an actual first-instability path. Do not infer Hopf from dominance alone.

2.4 T4: large-network selected-wave-number limit — RED

Exact published statement?
S03 Proposition 6 establishes an analogous explicit limit for Lorenz–96; a DM-specific limit has not been certified.

Immediate corollary?
For a fixed continuous symbol with a unique maximizer, convergence of grid maximizers is generic. Multiple maximizers and size-dependent parameters require different statements.

Routine supporting specialization?
Yes for the basic ratio limit.

Known method, new closed form?
At most a new coefficient substitution.

Substantive model-specific novelty?
Not the limit alone. Finite-size instability consequences may be useful as part of the core classification.

Likely referee response?
Dense sampling does not itself justify a principal asymptotic theorem.

Required sharpening.

P06 proposal S06: quantitative uniform error for both selected angle and first threshold under explicit nondegeneracy, with boundaries of applicability identified. Even those estimates need interpretive value; adding an error order alone does not earn GREEN.

2.5 T5: exact matched-bilinear comparison — YELLOW

Exact published statement?

MB at $n = 5, q = 2$ is exactly contained in S05. The intended comparison with DM was not found in the inspected corpus; full priority remains dependent on T1 and the enzymatic sequence.

Immediate corollary?

MB formulas follow familiar cyclic polynomial analysis. A nonempty region with a specified qualitative contrast is not supplied merely by the comparator's definition.

Routine supporting specialization?

Separate equilibrium and spectrum tables would be modest supporting work.

Known method, new closed form?

Yes, potentially a new comparative specialization. YELLOW is conditional and does not certify originality.

Substantive model-specific novelty?

Possible after a contrast theorem, not presently demonstrated. Equal parameter count and graph do not ensure equal equilibrium, response scale or Jacobian signs.

Likely referee response?

Is the contrast robust or an arbitrary consequence of comparing differently calibrated regimes?

Required sharpening.

P08 proposal S08: freeze the comparison rule and classify a nonempty open regime with a genuine mode, bifurcation-type, or criticality distinction; delimit existence of both branches.

2.6 T6: first Lyapunov coefficient — ORANGE

Exact published statement?

S02 and S08 publish Fourier reductions; S02 already includes cubic contributions. No DM coefficient is verified here.

Immediate corollary?

The contraction/selection-rule method is standard once Taylor tensors, inverses and a simple Hopf setting are justified.

Routine supporting specialization?

A long substituted formula is usually supporting calculation, even if algebraically laborious.

Known method, new closed form?

This is the most defensible description of a future DM formula; its closed-form priority is not established.

Substantive model-specific novelty?

A uniform or sharply delimited criticality classification may be substantive. No sign theorem is supplied.

Likely referee response?

What does the formula decide that the general Hopf machinery or a single numerical evaluation does not?

Required sharpening.

P13 proposal S13: rigorous sign/nonvanishing on a meaningful positive parameter region, all selected modes covered by the claimed range, and explicit exclusions. Do not promise a Bautin point without its further nondegeneracy conditions.

2.7 T7: heterogeneous Fourier Green function — RED

Exact published statement?

The general inverse-circulant identity is known (S07). A DM forcing numerator may be model-specific.

Immediate corollary?

The linear response follows differentiation of a nonsingular equilibrium equation and diagonal inversion.

Routine supporting specialization?

Yes; localized forcing does not automatically imply quantitatively localized response.

Known method, new closed form?

Yes for the coefficient specialization; that alone has little novelty. S19 supplies a stronger decay baseline.

Substantive model-specific novelty?

Only a sharpened directional, uniform or critical-scaling result with controlled nonlinear error is a candidate. It must improve on, not merely restate, generic decay theory.

Likely referee response?

Why is this more than an inverse DFT?

Required sharpening.

P16 proposal S16. Specify graph distance, parameter family and constants independent of size where claimed. Distinguish static sensitivity near zero eigenvalues from resonant harmonic response near a nonzero-frequency Hopf point.

2.8 T8: robustness bounds and threshold shifts — RED

Exact published statement?

Generic matrix-gap, resolvent and simple eigenvalue derivative results are published (S10–S11/S21); the specific nonlinear parameter certificate is not established here.

Immediate corollary?

The unstructured normal-matrix estimate and a first-order shift at a simple transverse crossing are routine applications.

Routine supporting specialization?

Yes for a loose sufficient bound or a left/right eigenvector derivative without further consequences.

Known method, new closed form?

Possibly a useful parameter-level formula, provided equilibrium displacement and remainder are controlled.

Substantive model-specific novelty?

A sharp certificate for an actual coefficient uncertainty class is a candidate; no such result is yet verified.

Likely referee response?

Is this just a spectral margin divided by a rough Lipschitz constant? Is the purported pattern effect nonzero under the imposed zero-mean constraints?

Required sharpening.

P18 proposal S18: explicit structured parameter radius with positivity/local existence and equilibrium motion included; quantify conservatism or sharp scaling. Compare any disorder-stabilization or second-order mode-mixing claim against S24. Do not identify a general full-block stability radius with a repeated/diagonal coefficient radius.

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